A General Multi-Source Data Fusion Framework

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ABSTRACT
With the development of the Internet, the increase of information sources and speed of information release and transmission have led to a sharp increase in the amount of information. To enable users finding more accurate and reliable information in the large heterogeneous multi-source data, data fusion technology becomes more and more important. Data fusion technology structuralizes and integrates heterogeneous data from different sources which greatly improves the comprehensiveness, availability and extensibility of data. This paper proposes a general multi-source data fusion framework. The framework transforms multi-source structured data, semi-structured data and unstructured data into unified data format described by RDF (Resource Description Framework) standard, and then realizes information fusion through data fusion algorithm, to solve the heterogeneity and semantic conflict in multi-source data fusion under the big data environment.

CCS Concepts
• Information systems→Data management systems→Information integration→Mediators and data integration

Keywords
multi-source data; data fusion; RDF; data matching; conflict resolution.

1. INTRODUCTION
The concept of data fusion was first applied in the military field. It refers to the process of combing data and information form multi-sensor together to obtain more accurate position and identity estimation, to realize the integrity evaluation of the battlefield situation [1].

Data fusion can combine data from different sources and provide uses with a unified data interface [2]. This paper presents a general framework for multi-source data fusion based on RDF. RDF (Resource Description Framework) is a basic ontology description language that provides a common data model to support the description of Web resources. RDF defines a complete set of systematic terms for specifying the parts of a resource. The part used to describe the object ontology is called the subject of a RDF description, the part used to describe the different features of the subject is called predicate, and the part used to express the feature values is called object. RDF defines a general framework, "subject-predicate-object" (<S, P, O>) triples to represent variety resource.

As shown in Figure 1, the proposed framework consists of four layers: input layer, data formatting layer, fusion layer and output layer. The input layer is multi-source structured data, semi-structured data and unstructured data. The data formatting layer transforms the data of input layer into a unified data format, which describing in RDF. And then formulates the mapping rules of partial ontology to global ontology. The fusion layer uses data fusion algorithm to fuse data from different sources. The output layer gets the final fusion result and provides users with more complete information.

This paper proposes a general four-layer data fusion framework to solve heterogeneous and semantic conflicts in multi-source data fusion. The organizational structure of this paper is as follows: Sec 1 introduces the four-layer multi-source information. Sec 2 introduces the related work. Sec 3 elaborates each part of information fusion framework in detail. In sec 4, an example is given to prove the feasibility of the information fusion framework. And sec 5 gives a summarizes of the whole paper.

2. RELATED WORK
In recent years, domestic and foreign scholars have done a lot of research on the framework of data fusion. Xu et al. [3] designed an ontology-based knowledge fusion framework to reduce the scale of fusion knowledge and improve its effectiveness and accuracy. Xie [4] designed a knowledge fusion framework based on the agricultural ontology and fusion rules, and divided the fusion process into three types: attribute fusion, instance fusion and concept fusion. Liu et al [5] defines a structure of multi-domain ontology, which provides need-based dynamic ontology for knowledge fusion through ontology mapping between different

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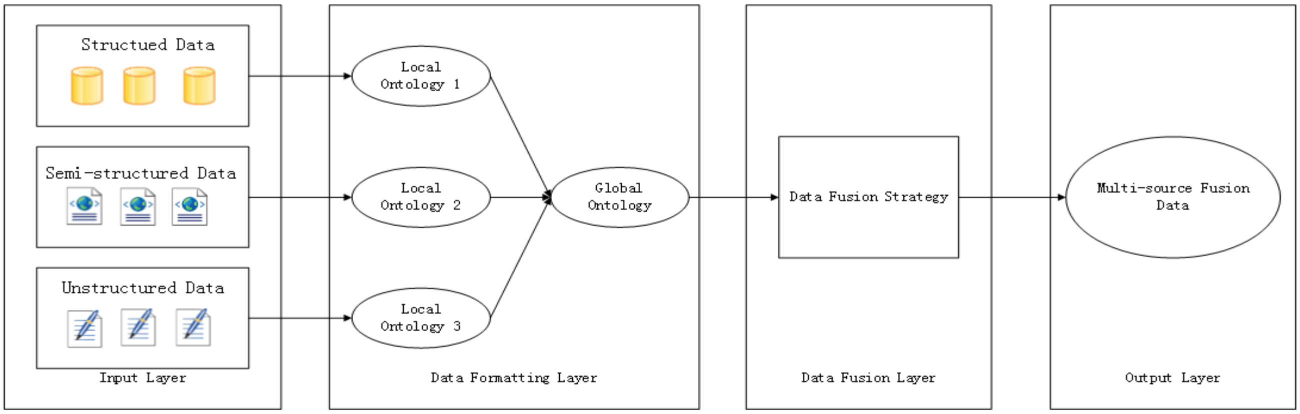

Figure 1. Multi-source information fusion framework

domains. Yu et al [6] proposed a knowledge-based 3D visualized Web service environment, encapsulated Web services, knowledge fusion process and knowledge base management in the application interface, and realized the construction of knowledge service model and knowledge driven service system. These studies discuss the concept of fusion and the functional modules of fusion process, but lack universality and difficult to realize. In this paper, a data fusion framework with universality and extensibility is proposed.

3. DATA FUSION FRAMEWORK

Data from different sources is distributed, heterogeneous. In addition, it also has the characteristics of redundancy, noise, uncertainty and incomplete. Data cleaning cannot completely solve these problems, so form the perspective of data source, it is necessary to fuse data from different sources into a unified representation.

3.1 Data Formatting and Ontology Mapping

The structured data used in this article is derived from relational database. According to the mapping rules of relational database to RDF data proposed in previous work [7], R2RML (RDB to RDF Mapping Language) mapping tool is used to complete the transformation of data from relational database to RDF data. Semi-structured data and structured data are extracted from online encyclopedias, blogs and other Web sites according to certain rules. All these data are stored as RDF format.

Definitions of concept and predication generally do not conflict in the same data source. However, when data from multiple sources, there are some similar concepts and predications. They may express the same meaning or they may express different meanings. Therefore, before data query and match process, similarity detection rules are used to determine whether they are the name. The similarity methods used in this paper include conceptual similarity, attribute similarity, relational similarity and data format similarity. After a series of similarity detection methods, we map local ontology to global ontology through merging and conflict resolution operations. Merging refers to mapping the same or similar local ontology which express the same meaning in different data sources to the global ontology and eliminate redundancy. Conflict resolution refers to mapping the same or similar local ontology which express different meanings to the global ontology and distinguishing them to resolve conflicts. In addition to similarity-based approach, the ontology mapping process can also use manually defined methods. According to the expert knowledge, map local ontology to the global ontology.

3.2 Data Query and Match

It is very important to locate exactly where the fusion operation id needed in the process of data fusion. A good query and match algorithm can improve the efficiency and accuracy of data fusion.This section describes the query and match algorithm used in this paper.

RDF data can be represented by a graph model. An RDF data graph, denoted as $G = (V, E, L)$, represent the relationships between RDF triples "subject-predicate-object", where V is a set of vertexes, E is a set of edges between vertexes and L is a set of labels. V corresponds to subject and object in triples. E corresponds to predicate in triples. $L = L_v + L_p$, where L_v is a set of vertexes labels and L_p is a set of edges labels. Based on RDF graph model, data query processing can be transformed into a subgraph matching problem on a large graph. In order to improve the query efficiency, the RDF graph is divided by clustering method in this paper, and then a tree-based index is established according to the result of partitioning [8]. When performing the query operation, the query subgraph is matched according to the index to reduce the search space at first, and then find the corresponding query results on the reduced subgraph.

First, we cluster the vertexes in a RDF source by subject. The clustering process follows the principle that each two classes cannot intersect each other. Then calculate g radius and the center point after clustering. We can construct a balanced binary tree index according the clustering results. Because leaf nodes are the result of clustering in RDF graphs that are known already, a bottom-up approach is used to generate tree structures. The generation of non-leaf nodes is made from a leaf node and a node closest to the leaf node, which generates a parent node of two nodes. The value of the parent node is the radius and center of the two nodes. Repeat this process until the balanced binary tree contains all vertexes in the RDF source. Since the selection of leaf nodes is random, the final balanced binary tree is not unique at each constructing time. Each node in the binary tree represents a set of elements in the RDF source, in other words, a set of nodes that have been clustered. When the distance between the query data and a node in the balanced binary tree is less than the radius of the node in the balanced binary tree, it can be determined that the data's looking up range. In the process of query, it is equivalent to look up element in subgraph, which reduces the searching space and improve efficiency of searching.

After establishing the tree index, we begin query from the root node. Calculate the distance between the matching data and the left and the right child nodes respectively, select the subtree with small distance as the new start node of next query. Repeat this step again and again until reach a leaf node. This leaf node does not represent the final match result. In the process of fusion, there are cases where the subject is the same or similar but represent different meaning. Therefore, after we matching a leaf node, we should also calculate their object similarity. When their object similarity is greater than a certain threshold value, we consider the match successful. Otherwise, the match is unsuccessful.

We summarize the data fusion query and match procedure in Algorithm 1. In the algorithm, the input RDF data resource. If match successfully, it means two resource from different source are similar and they can be fused. Match result is the fusion location. Otherwise, it means there is no similar resource in the original RDF data source, we do add operation with new resources.

Algorithm 1: Query and match Procedure

Input: RDF data resource

Output: Query and match result

1) Cluster nodes in RDF graph by subject

2) Construct a balanced binary tree index according the clustering results

3) Calculate the distance between query data and the nodes in the balanced binary tree, determine the search space

4) Query by subject similarity within the range determined in 3)

5) If subject match successfully, then queries by object similarity

6) Subject and object all match successfully, return fusion location; otherwise return null.

3.3 Conflict Resolution in Data Fusion

After discovering the location where data fusion is needed, we do data fusion operations according the fusion strategy. In the previous work [7], two kinds of conflicts were solved according to practical application requirements. One is multi-value of subject conflict and the other one is multi-value of object. The solution proposed for the two conflicts is to achieve the fusion results through syntactic fusion, that is, logical addition operation with multi-value of subject or object [9]. The calculation method is shown in formula (1) and formula (2).

$$\text{Syntax}(\Sigma) = \text{Syntax}(s_1, s_2 \dots s_m) \quad (1)$$

$$\text{Syntax}(\Delta) = \text{Syntax}(o_1, o_2 \dots o_m) \quad (2)$$

where Σ represent the set of subjects and Δ represent the set of object. Syntactic fusion method is only a preliminary solution to multi-value subject and object, eliminating redundant items in the subject set or the object set. Clearly, the granularity of this solution is not small enough. There may be contradictory terms between multiple subject or object values. Syntactic fusion method is meaningless to this problem. In the specific fusion process, we should make a judgment on the authenticity of the data according to the differences characteristics of the data. By analyzing the description of the resource, we should find the most relevant values as the result of fusion.

Algorithm 2: Conflict resolution based on word vector

Input: RDF triples <S, P, O>

Output: k-dimension word vector

1) Use Word2vec method to calculate vector representation of triples.

2) Calculate the cosine similarity to measure the similarity between the attributes.

3) If the similarity exceeds a certain threshold, we consider they express the same meaning and keep either of the two attributes. Otherwise, the meaning of the two attributes are different, we should keep both attributes.

In the fusion process, since the syntactic fusion algorithm is not applicable to all conflict types, the additional known information should be fully considered. We should combine semantic information to solve the value confliction on object. In this paper, we adopt the method based on word vector to achieve better fusion effect. Word2Vec [10] is an open source tool released by Google in 2013. It transforms text content into a vector in the k-dimensional vector space. The similarity of two texts is obtained by calculating the similarity of the k-dimension vectors of two words. The detailed algorithm flow description is shown in Algorithm 2. The basic idea is to compute the embedding vectors of triples <S, P, O> in RDF files. After we obtain the embedding vector, we calculate the similarity between the attributes that need the fusion operation. If the similarity exceeds a certain threshold, we consider they express the same meaning and keep either of the two attributes. Otherwise, the meaning of the two attributes are different, we should keep both attributes.

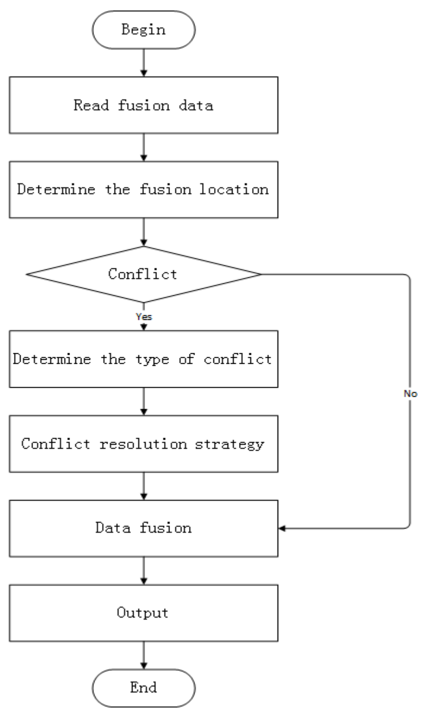

Figure 2. Data fusion process graph

3.4 The Overall Framework of Data Fusion

The overall framework of data fusion is divided into three steps:

1) Data Formatting and Ontology Mapping Data query and match.
2) Data Query and match.
3) Conflict resolution in data fusion.

The three subsections detail the algorithms and workarounds used in these three steps. The complete data fusion process is described as follows and the data fusion process graph is shown in Figure 2

1) Transform structured data, semi-structured data and unstructured data from different sources into RDF format and map local ontology to global ontology.
2) Use the data query and match algorithm to locate the location where the data need to be fused, and then mark the location in the original RDF resource.
3) Determine whether there is a conflict, if has, judge conflict type and resolve the conflict according to the conflict type.
4) Fusion data from different sources and record the match numbers of the relevant data.
5) Save the fused data into a new RDF resource.

4. APPLICATION SCENARIO ANALYSIS OF THE DATA FUSION FRAMEWORK BASED ON RDF

The purpose of data fusion is to combine heterogeneous data from multiple sources and multiple domains to form a unified entirety., which is convenient for data users to quickly obtain more extensive and comprehensive data information. In this paper, the paper information data and college information data are used in the research. We use the framework proposed in this paper to integrate these two types of data. The data model diagram of papers and college before fusion are shown in Figure 3(a)(b). The data model diagram after fusion is shown in Figure 3(c).

The authors' organizations in the paper data and the college's names in the college data are fused into a new data model set "Organization[Name]" through data fusion framework. After data fusion operation, by inputting a paper or an author name, users can quickly obtain the paper publication information, which authors are from the same institution. Figure 3 only shows the connection effect of the "Organization[Name]" data model. Similarly, it can also fuse the field that papers belong to and field where experts and scholars research to obtain the field data model. So that users can quickly obtain relevant papers' and experts' information when query a certain paper.

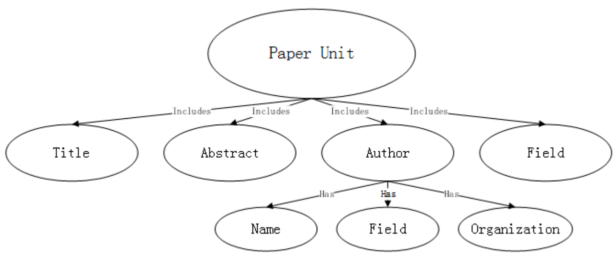

Figure 3(a). Paper data model

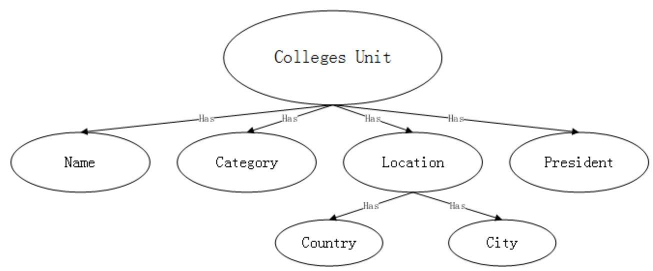

Figure 3(b). College data model

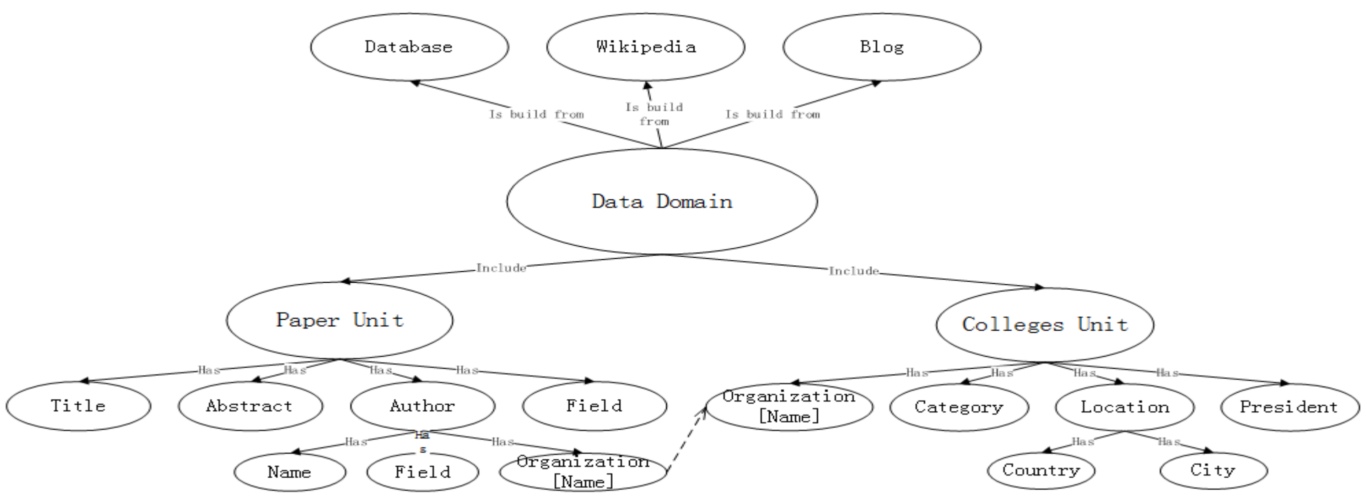

Figure 3(c). Paper and college fused data model

5. CONCLUSION

This paper improves the original RDF based data fusion framework, optimizes the methods of data query and match, matches data within the scope of the subgraph. It not only speeds up data query and match, but also reduces memory usage. At the same time, a suitable solution is proposed for processing the fusion conflict that cannot be solved in our previous framework. However, the current data fusion framework is based on the same language, that means the framework cannot be applied to the multi-language data fusion environment. Considering that the actual application may involve multi-language data fusion scenarios, we will work on this issue in the follow up works.

6. ACKNOWLEDGMENTS

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